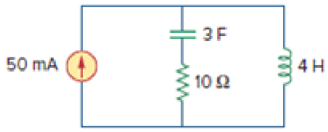
Chapter 8, Problem 14PP

Problem

Draw the dual circuit of the one in Fig. 8.46.

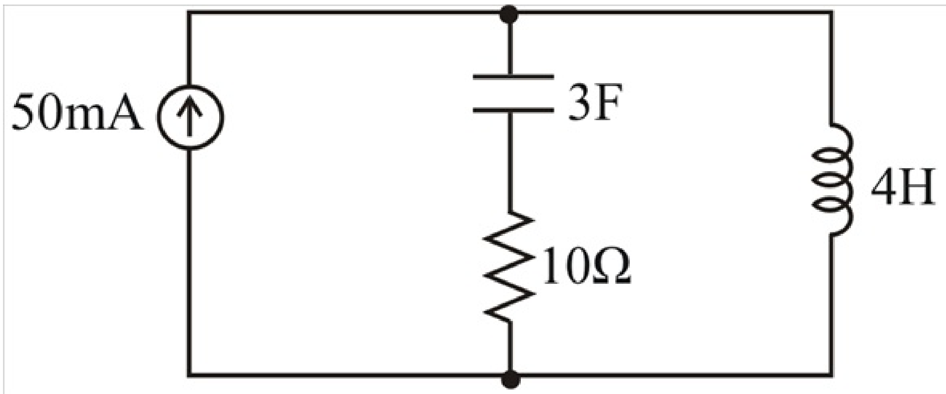
Figure 8.46:



Step-by-step solution

Step 1 of 5

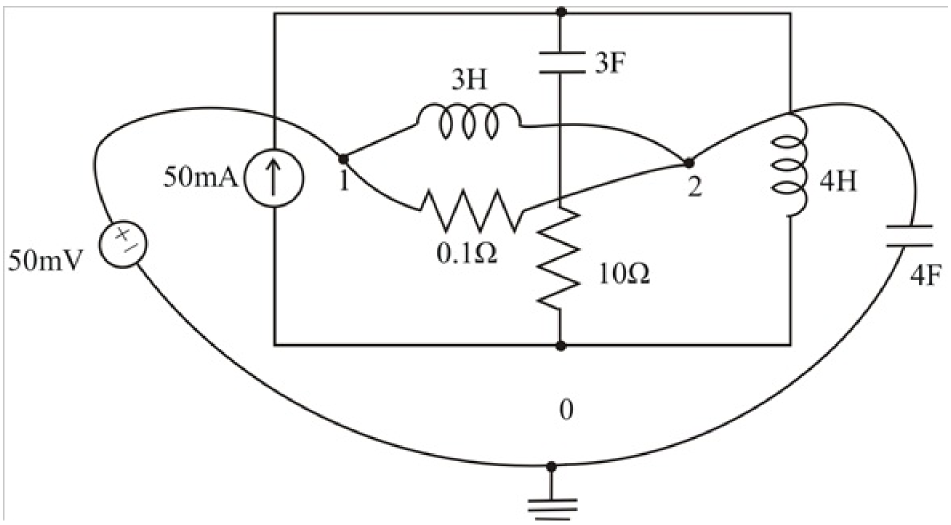
The given circuit is



Step 2 of 5

We first locate nodes 1 and 2 in the two meshes and also the ground node 0 for dual circuit. We draw a line between one node and another crossing an element. We replace the line joining the nodes by the duals of the elements which it crosses.

Step 3 of 5



Step 4 of 5

For example, the line between nodes 1 and 2 crosses a 3F capacitor and 10Ω resistor, therefore they are replaced by their duals. 3F capacitor with 3H inductor and 10Ω resistor with $\frac{1}{10}\Omega$ or 0.1Ω resistor. Since they are in series, in the dual circuit they are in parallel.

A line between nodes 1 and 0 crossing the 50 mA current source will contain a 50 mV voltage source. By drawing the lines crossing all the elements, we construct the dual circuit on the given circuit as shown above.

Step 5 of 5

Hence the dual circuit is as shown below,